

# Lakshay Chauhan

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## EDUCATION

**Indraprastha Institute of Information Technology (IIIT) Delhi**

Jul 2021 – Jun 2025

*Bachelor of Technology in Computer Science and Engineering*

*Delhi, India*

## WORK EXPERIENCE

**Member of Technical Staff**

Jul 2025 – Present

*ZL Technologies, Inc* | [Website](#)

*Hyderabad, India*

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

**Research Assistant** | [Gitlab Project Organization](#)

May 2024 – May 2025

*Networks and Systems Security Lab - IIITD* | [Website](#)

*Delhi, India*

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

## PROJECTS

**Minimal Compiler Infrastructure** | [Repository](#)

May 2024 – Present

- **Key Skills:** *Compiler Infrastructure, Design and Optimization, Rust, FASM, WASM, Graphviz*
- Developed a minimal compiler infrastructure inspired by LLVM. It converts source code into an optimized control flow graph (CFG) and translates it into target-specific assembly code.
- Implemented CFG optimization passes, such as identifier validation, constant folding, and CFG simplification, ensuring high-performance code generation.

**Immediate-Mode GUI Framework** | [Repository](#)

Feb 2024 – May 2024

- **Key Skills:** *C/C++, Graphics Programming, Systems Programming, UI Frameworks, Rendering Pipelines*
- Designed and implemented a lightweight immediate-mode GUI framework from scratch using a low-level graphics library. Built a real-time rendering loop, input/event handling system, and modular UI components with minimal abstraction overhead.
- Focused on performance, explicit state management, and predictable memory usage for interactive applications.

**Needham Shcroeder Based PDF Print Server** | [Repository](#)

Mar 2025 – May 2025

- **Key Skills:** *C/C++, Networking, Cryptography, Client-Server Systems, Security Protocols*
- Implemented a secure client-server print service using the Needham-Schroeder authentication protocol. Designed custom request/response messaging and session validation to prevent unauthorized access. Gained hands-on experience with low-level networking, protocol correctness, and security-aware systems design.

## TECHNICAL SKILLS

**Languages:** C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell

**Frameworks:** PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib

**Developer Tools:** Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SQLite3

**Technical Electives:** Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing

## AWARDS

- Awarded the **Summer Undergraduate Research Fellowship** (2023) by **IRD-IIITD** for the project “Utilizing Ultrasonic Distance Sensors as a Mapping Tool to Design User-Friendly CST”.
- Awarded the **CHANAKYA Fellowship** (2024) by **iHub Anubhuti Foundation** for the project “A Unified Approach to User Emotion Detection through Emojis and Textual Analysis”.